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Machine Learning-Driven Seismic Facies Analysis for Channel Prospect Identification in Offshore Abu Dhabi

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Abstract

This paper presents an integrated approach to seismic facies classification for characterizing channel development in the H carbonate formations of offshore Abu Dhabi. Focusing specifically on waveform classification and multi-attribute analysis methods, we demonstrate how these techniques significantly enhance the identification and interpretation of subtle channel features that conventional methods often fail to detect. Our workflow combines unsupervised classification using SelfOrganizing Maps (SOM), supervised neural network classification, and hierarchical waveform classification to analyze the complex channel systems within these challenging carbonate reservoirs. Results show that the integration of these methods provides robust verification of channel features and deeper insights into their genetic mechanisms, including tectonic influences, sea-level fluctuations, karstification processes, and depositional controls. Validation against RGB color blending results by decomposed seismic data demonstrates much better classification accuracy and much more obtained details, significantly outperforming conventional amplitude-based interpretation. This approach offers valuable insights for optimizing development strategies in similar carbonate channel systems throughout the region.

Introduction

Channel systems within the H carbonate formations of offshore Abu Dhabi represent significant exploration and development targets, yet they present unique challenges for seismic interpretation due to their subtle expression and complex genesis. Core-based study demonstrates a depositional environment extending from intertidal to outer ramp settings in a carbonate ramp system^[1]. Within this system, tidal channels commonly develop porous facies^[2-6], making them effective reservoir units. Traditional amplitude-based interpretation approaches often fail to fully characterize these features, leading to incomplete understanding of reservoir architecture and connectivity.

The H Formation forms part of the Upper Cretaceous (Santonian-Campanian) Aruma Group in offshore Abu Dhabi, representing a significant carbonate platform development during a dynamic period of regional tectonic activity. The formation consists predominantly of fine-grained chalky carbonates deposited in a shallow marine environment, with varying degrees of influence from the tectonic activity associated with

the emplacement of the Semail Ophiolite. Channel features within the H Formation have been recognized in previous studies but remain poorly understood in terms of their genesis, distribution, and reservoir implications^[1-6].

Recent advances in waveform classification and multi-attribute analysis offer new opportunities to enhance the identification and characterization of these subtle channel features^[7-10]. These techniques can identify patterns and relationships in seismic data that may not be apparent through conventional analysis, providing crucial insights for reservoir characterization and development planning.

This paper presents an integrated workflow that combines multiple classification approaches:

- Waveform classification (both unsupervised and supervised)
- Multi- attribute analysis using Self-Organizing Maps (SOM)
- Principal Component Analysis (PCA) for optimal attribute selection
- Neural network classification for supervised facies prediction

By applying this multi-method approach to the H formation channel systems, we aim to:

- Enhance the visualization and characterization of channel features
- Verify the presence and extent of these features through independent methods
- Determine the primary causal mechanisms for channel development
- Establish a comprehensive model for the evolution of these features

The findings of this study have significant implications for understanding the stratigraphic architecture and reservoir characteristics of the H Formation and provide insights applicable to similar carbonate settings throughout the region.

Methodology

This section details the integrated workflow employed for machine learning-based seismic facies classification and presents the key results obtained from its application to the H Formation. The methodology combines various seismic analysis techniques to identify and characterize paleochannels, providing a robust framework for channel prospect identification.

Dataset

This study utilized a comprehensive 3D mega seismic dataset of full offshore Abu Dhabi for the initial study and a selected field covering approximately 1800 km², with a focus on the H Formation interval. The seismic data have the following characteristics: Inline/crossline spacing: 12.5 m, Sample rate: 2 ms, Dominant frequency: 30-40 Hz, Processing sequence: Pre-stack time migration with amplitude preservation.

Workflow Overview

The workflow for machine learning-based seismic facies classification is a multi-step process designed to progressively refine the analysis from a regional scale to a detailed local investigation. The steps are as follows:

1. Data QC: A fast seismic data QC is recommended for seismic facies analysis such as spectrum analysis, SNR, attributes extraction and display, etc.

2. **Initial Surface Picking:** A quick surface of the top of the H formation is picked for the entire merged offshore mega seismic volume. This provides a foundational structural context for subsequent analyses.
3. **Regional Waveform Classification:** Waveform classification is performed for the reservoir of H formation across the full merged offshore mega seismic volume. This initial classification helps in broadly identifying areas of interest based on seismic waveform characteristics.
4. **Area Selection for Detailed Study:** Based on the regional waveform classification, the most promising area for detailed study is selected. This focuses the subsequent, more intensive analyses on zones with higher potential for paleochannel development.
5. **Refined Surface Interpretation:** For the selected local area, the surface interpretation is refined. This ensures high-resolution structural control for the detailed facies classification.
6. **Machine Learning-Based Seismic Facies Classification:** Within the selected local area, machine learning-based seismic facies classification is conducted. This involves:
 - **Waveform Classification:** Further detailed waveform analysis.
 - **Attribute Extraction and PCA Analysis:** Extraction of various seismic attributes and application of Principal Component Analysis (PCA) to select optimal attributes that best represent geological features and reduce redundancy.
 - **Neural Network Analysis:** Application of neural network analysis. This can be supervised if well logs and stratigraphic interpretations are available for calibration, or unsupervised if no well data is available.
7. **Paleochannel Distribution:** The primary objective of the classification is to obtain the distribution of paleochannels.
8. **Verification through Spectral Decomposition and RGB Blending:** The identified paleochannel distributions are verified using spectral decomposition and RGB blending techniques. These methods provide visual confirmation and enhance the interpretability of the results.
9. **Morphological Verification (Optional):** If the data quality is sufficiently high, the morphology of the picked horizons can also be used to help locate or verify the paleochannel distribution^[10-11].

Waveform Classification Methodology

Waveform classification was applied to categorize seismic traces based on their shape and character within the H Formation interval. This technique provides an independent method for identifying and characterizing channel features based on their distinctive seismic response.

Unsupervised Waveform Classification. The unsupervised classification workflow included:

- Definition of analysis window encompassing the H Formation
- Application of neural network classification to identify natural groupings of waveforms (Fig. 1)
- Generation of 3-category classification to broadly differentiate channel, non-channel, and transitional facies
- Refinement through hierarchical classification to enhance detail within each primary category

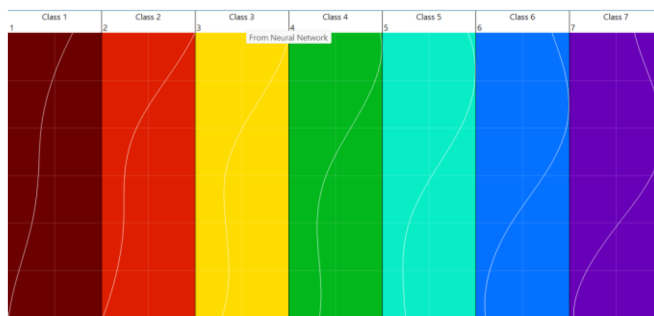


Figure 1—Model waveforms of unsupervised classification

This approach allowed for the identification of natural patterns in the data without imposing preconceived interpretations, providing an objective basis for channel identification.

Supervised Waveform Classification. The supervised classification workflow consisted of:

- Extraction of model waveforms from well locations with known lithofacies to replace the model traces (Fig. 2)
- Application of neural network classification using model waveforms
- Correlation of classification results with lithofacies from well data
- Generation of probability volumes indicating likelihood of specific facies

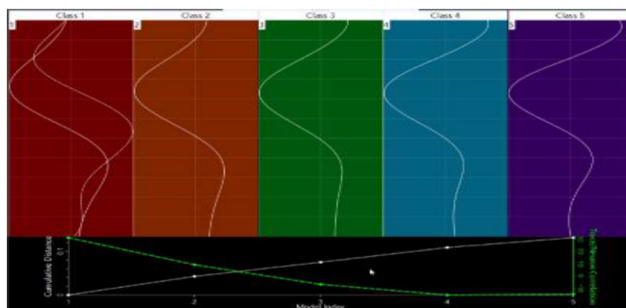


Figure 2—Model waveforms of supervised classification

This approach leveraged available well control to guide the classification process, enhancing the geological relevance of the results.

Multi-Attribute Analysis Methodology

Multi-attribute analysis was employed to integrate multiple seismic attributes and enhance the characterization of channel features.

Attribute Generation and Selection. A suite of seismic attributes sensitive to stratigraphic and structural features was generated: Instantaneous attributes (amplitude, phase, frequency), Geometric attributes (coherence, curvature, dip), Stratigraphic attributes (relative acoustic impedance, reflection strength), Spectral decomposition components (20Hz, 30Hz, 40Hz).

Principal Component Analysis (PCA) was applied to identify attributes with maximum variance and eliminate redundancy^[12,13]. The analysis identified five key attributes contributing maximum variance: spectral decomposition components (20Hz, 30Hz), coherence, most positive curvature, and relative acoustic impedance (Fig.3).

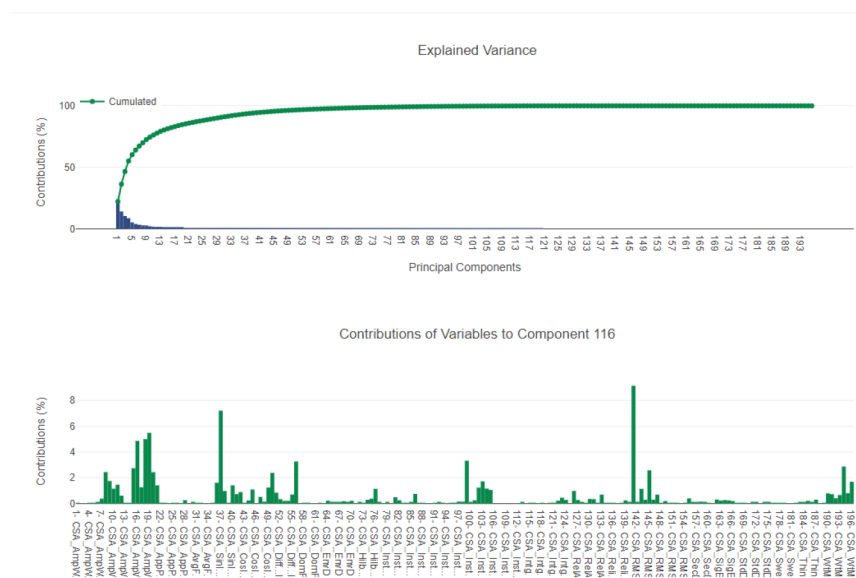


Figure 3—PCA analysis of multiple attributes

Self-Organizing Maps (SOM). Self-Organizing Maps were applied to the selected attributes following this workflow:

- Normalization of selected attributes to ensure equal weighting
- Application of SOM with varying neuron counts (25, 64, 100) to identify optimal resolution
- Training of the SOM using competitive learning algorithms
- Mapping of SOM results to 2D color maps for visualization
- Interpretation of SOM clusters in terms of geological features

The SOM analysis with 64 neurons provided the optimal balance between detail and interpretability^[14], revealing distinct channel facies patterns not apparent in conventional displays.

Integrated Interpretation Workflow

The results from waveform classification and multi-attribute analysis were integrated through a systematic workflow:

- Initial identification of channel features using individual methods
- Comparison of results from different classification approaches
- Integration of results to develop a consensus interpretation
- Correlation with well data to ground-truth seismic interpretations
- Development of a comprehensive model of channel development

This integrated approach ensured that channel features were identified and characterized using multiple independent methods, providing a robust foundation for interpreting their genesis and evolution.

Application and Results

This section presents the key results obtained from the application of the integrated machine learning workflow to the H Formation, demonstrating its effectiveness in identifying and characterizing paleochannels.

Waveform Classification Results

Unsupervised Classification. The unsupervised classification results showed: Clear differentiation between channel and non-channel facies, Enhanced definition of channel margins compared to amplitude displays, Identification of internal facies variations within channel features, Recognition of hierarchical relationships between channel systems.

Hierarchical classification further refined these results, revealing: Multiple generations of channel development, Variations in channel fill characteristics, Relationship between channels and surrounding facies.

The unsupervised approach was particularly effective at identifying the overall distribution and morphology of channel systems without bias from preconceived geological models^[13].

Supervised Classification. Supervised classification, calibrated to well data, demonstrated: Strong correlation between classified channel facies and core-described lithologies, Differentiation between erosional channels and dissolution-enhanced features, Identification of karstified zones associated with channel-like features, Correlation between waveform characteristics and specific depositional environments.

The supervised approach provided greater geological context for the classification results, enhancing their interpretive value for reservoir characterization^[14].

Multi-Attribute Analysis Results

The SOM analysis revealed distinct multi-attribute clusters associated with channel features: Channel margins were characterized by high coherence contrast and distinctive curvature signatures, Channel fill showed variable acoustic impedance patterns suggesting different fill lithologies, Karst-affected areas displayed characteristic combinations of attributes distinct from purely depositional channels, sweet spots within channels were identified by specific attribute combinations.

The multi-attribute results indicated that the channel features represented a complex interplay of depositional, structural, and diagenetic processes, rather than simple erosional or depositional channels.

Integration of Results

The integration of results from waveform classification and multi-attribute analysis provided a comprehensive characterization of channel features in the H Formation:

Waveform classification effectively identified the presence and general morphology of channel features, multi-attribute analysis revealed the complex nature of these features and their relationship to structural and diagenetic processes, Areas of agreement between methods provided high-confidence channel interpretations, Areas of disagreement highlighted zones requiring additional investigation.

The combined results demonstrated that what initially appeared as simple channels actually represented a complex system of features influenced by multiple genetic mechanisms.

Visualizations of Workflow Results

The application of this integrated workflow yielded significant insights into the distribution and characteristics of paleochannels within the H Formation. The results are visually represented through various figures, demonstrating the effectiveness of the machine learning approach.

Waveform Classification of the Full Offshore Abu Dhabi. Fig. 4 illustrates the waveform classification of the full offshore Abu Dhabi seismic volume. This regional view provides a broad understanding of the seismic facies distribution, highlighting areas with distinct waveform characteristics that may correspond to different geological features, including potential channel systems in the middle and east area of offshore Abu Dhabi. The varying colors represent different waveform classes, indicating variations in rock properties and fluid content across the vast area.

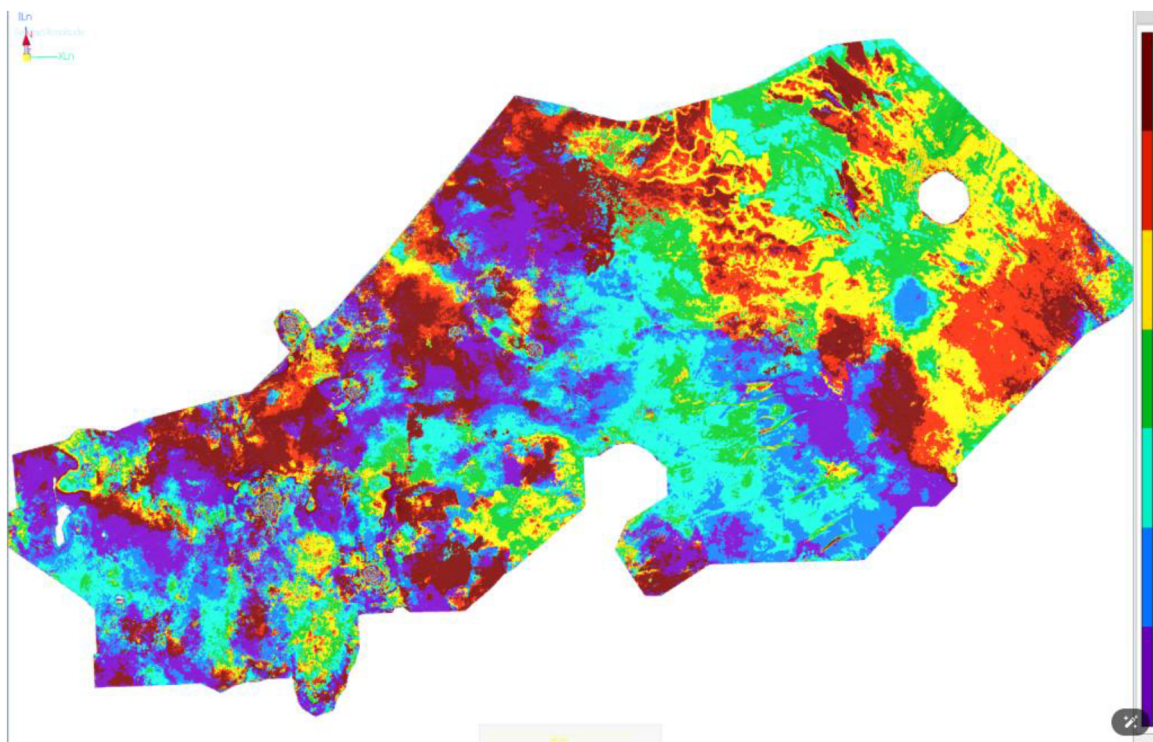


Figure 4—The waveform classification of the full offshore Abu Dhabi

Machine Learning-Based Facies Classification of a Selected Local Area. Fig. 5 presents the detailed machine learning-based facies classification for a selected local area in the middle of offshore Abu Dhabi. This zoomed-in view, derived from the refined analysis, reveals intricate patterns of seismic facies. The classification, informed by waveform analysis, attribute extraction, and neural networks, delineates specific facies that are indicative of paleochannels. The enhanced resolution in this local area allows for a more precise identification and mapping of these subtle features.

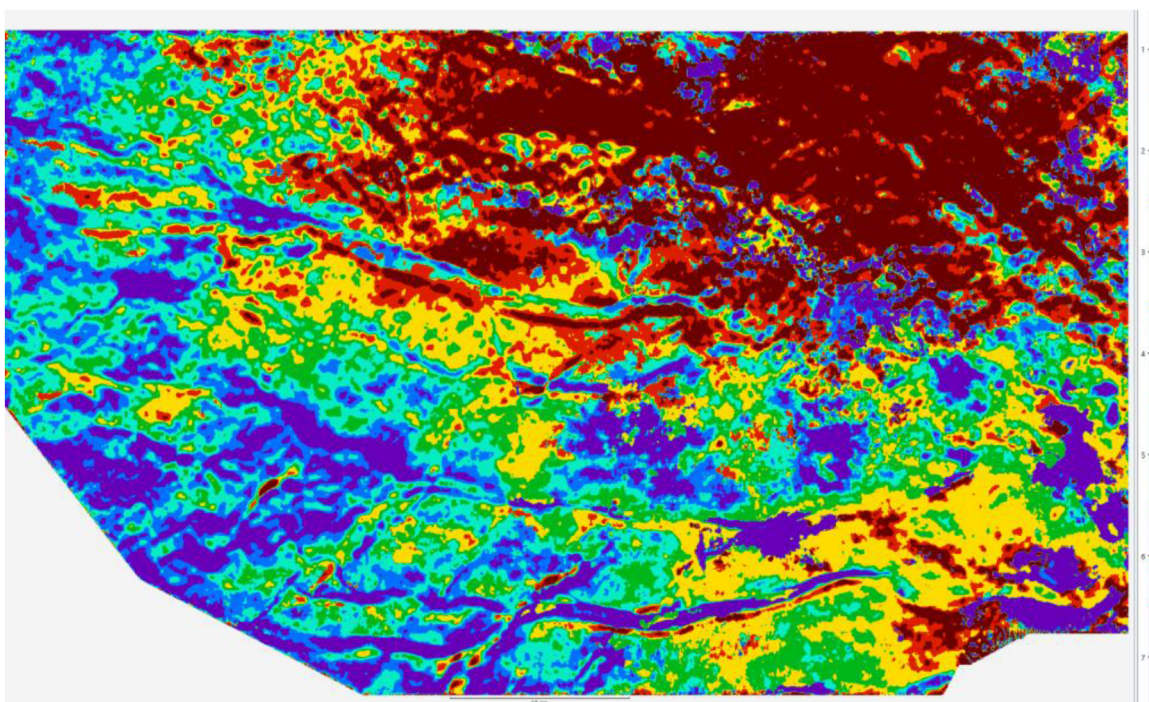


Figure 5—The machine learning based facies classification of a selected local area

RGB Blended Seismic Section Showing Perpendicular Cut of Paleochannels. Fig. 6 displays an RGB blended seismic section, providing a perpendicular cut through the identified paleochannels. This visualization technique, combining different seismic attributes (e.g., amplitude, frequency, phase) into red, green, and blue color channels, enhances the visibility of geological features. The distinct color anomalies and linear features observed in this section strongly indicate the presence and orientation of paleochannels, offering a cross-sectional view of their geometry and fill characteristics. The channel like anomaly is shown on the right-hand side of the section below the dashed line (H surface).

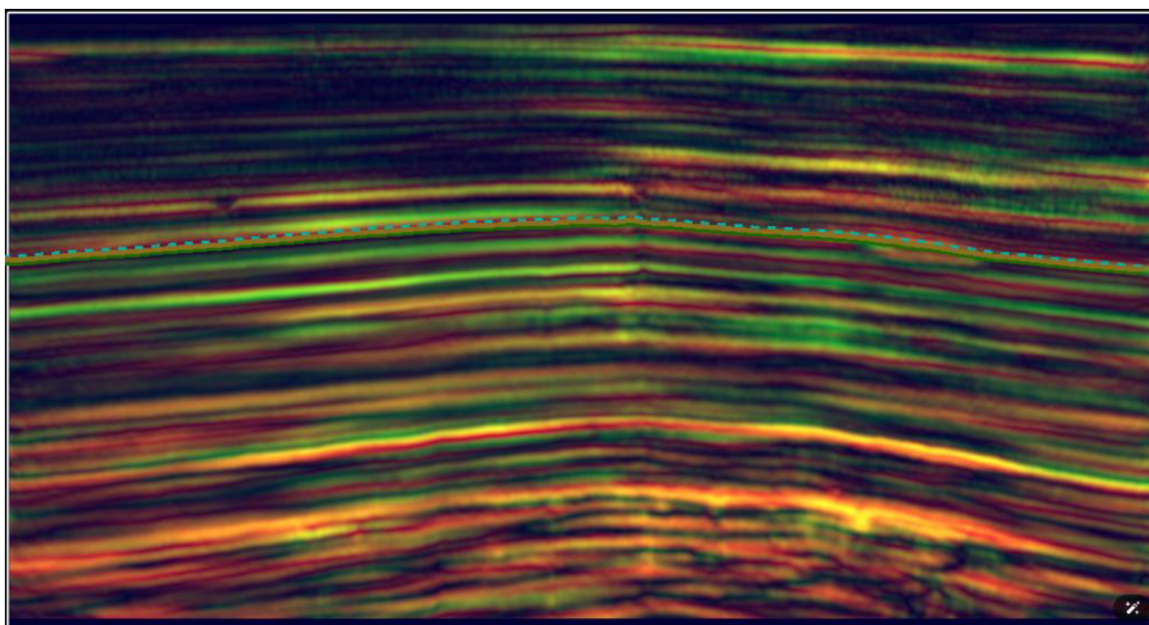


Figure 6—RGB blended seismic section showing the perpendicular cut of paleochannels

RGB Blended Maps Showing the Distribution of Paleochannels. Fig. 7 provides RGB blended maps that clearly show the areal distribution of paleochannels. These maps are crucial for understanding the spatial extent and connectivity of the channel systems. The patterns observed in these maps, characterized by distinct linear or curvilinear features, correspond to the paleochannels identified through the machine learning workflow.

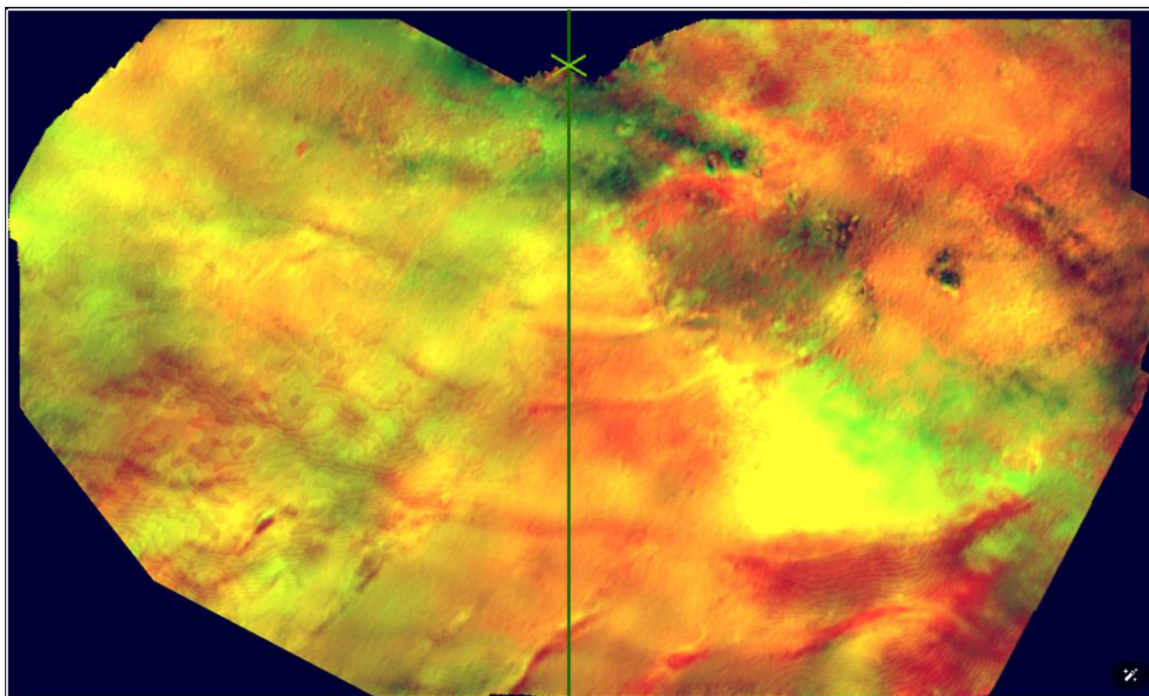


Figure 7—RGB blended maps showing the distribution of paleochannels

This comprehensive mapping is vital for prospect identification and reservoir characterization. These features are the verification of the seismic facies classification.

Refined Seismic Surface Showing the Distribution of Paleochannels. Fig. 8 illustrates the refined seismic surface, further highlighting the distribution of paleochannels. This image integrates the interpreted surface with the paleochannel mapping, providing a clear geological context. The refined surface, coupled with the paleochannel distribution, offers a comprehensive understanding of the subsurface architecture, which is essential for accurate drilling and development planning.

These results collectively demonstrate the efficacy of the machine learning-driven approach in identifying and characterizing complex paleochannel systems in challenging geological settings. The integration of various seismic attributes and advanced classification techniques provides a robust and detailed understanding of the subsurface, significantly enhancing channel prospect identification.

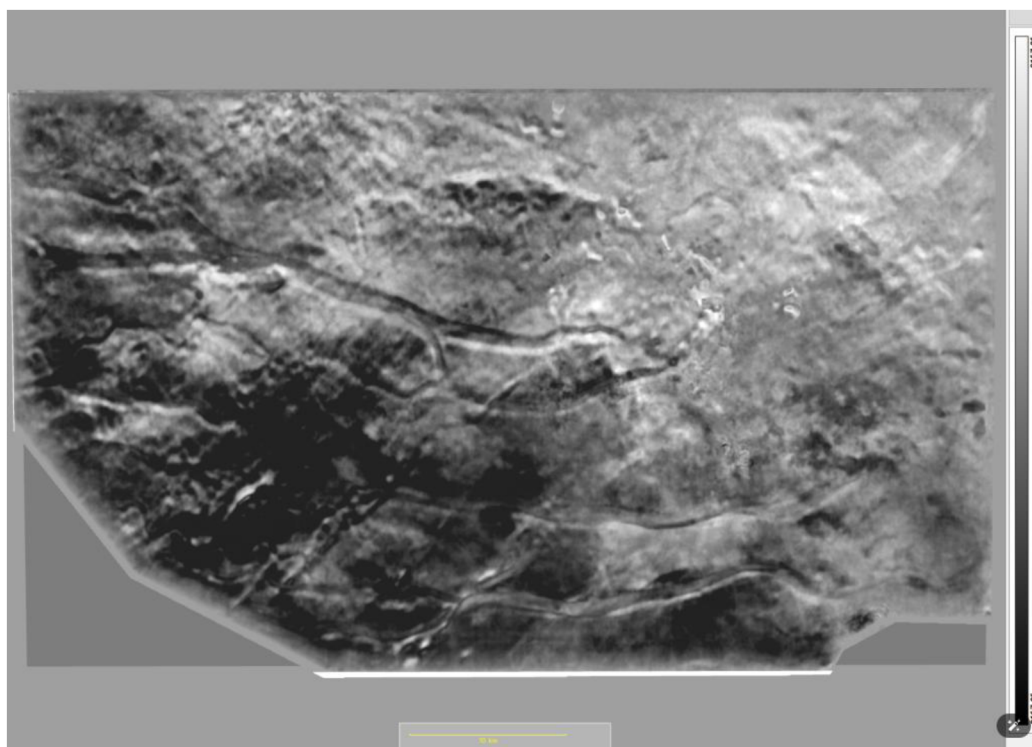


Figure 8—Refined seismic surface showing the distribution of paleochannels

Verification and QC. Blind well tests are commonly used for verification and QC. Data from core analysis, well log analysis, and stratigraphic interpretation can serve as tools for verification and quality control.

Discussion

The integrated analysis revealed that channel development in H Formation resulted from a complex interplay of multiple causal mechanisms:

Tectonic Influences. The emplacement of the Semail Ophiolite during the Late Cretaceous created significant tectonic influences on the H Formation: Forebulge development ahead of the advancing ophiolite created topographic highs susceptible to erosion^[1-6], Differential subsidence patterns established preferential pathways for channel development, Fault- controlled topography influenced drainage patterns and erosional processes, Angular truncations at the top H formation surface indicate tectonic tilting and erosion, Structural lineaments guided subsequent channel development and enhancement.

These tectonic influences established the primary framework for channel development, creating topographic variations that influenced subsequent processes.

Sea-Level Fluctuations. Late Cretaceous (Santonian-Campanian) sea-level fluctuations played a crucial role in channel development: regressions exposed portions of the carbonate platform to subaerial processes, subaerial exposure led to incision of channels through fluvial processes, subsequent transgressions modified these channels through submarine erosion, base level changes affected accommodation space and sedimentation patterns, similar patterns have been documented in the Natih Formation of Oman, providing a regional analog^[1-6].

The alternating periods of exposure and submergence created a complex history of channel incision, fill, and modification that is reflected in the seismic response.

Karstification Processes. The fine-grained chalky carbonates of the H Formation were particularly susceptible to karstification during periods of exposure: dissolution of carbonate material created and enhanced channel-like features, preferential weathering along fracture networks established karst drainage

systems, paleokarst development during exposure periods created complex dissolution networks, collapse features associated with karst processes created apparent channels in seismic data, Meteoric water infiltration enhanced dissolution along pre-existing weaknesses^[1-6].

Karstification processes both created new channel-like features and modified pre-existing channels, contributing to the complex nature of these features.

Depositional Controls. Original depositional patterns also influenced channel development: pre-existing depositional topography guided subsequent erosional processes, carbonate platform margin settings were particularly susceptible to channel development, sediment gravity flows and turbidites created channel-like features in some areas, contour currents along platform edges shaped elongate channel-like features, facies transitions affected erosion resistance and channel development^[1-6].

These depositional controls interacted with tectonic, sea-level, and karstification processes to produce the complex channel systems observed in the H Formation.

Advantages of Integrated Classification Approach

The integration of waveform classification and multi-attribute analysis provided several key advantages:

Complementary perspectives: Each method highlighted different aspects of channel features, providing a more comprehensive characterization.

Independent verification: Agreement between different methods increased confidence in the interpretation of channel features.

Enhanced discrimination: The combined approach improved discrimination between different types of channel features and their genetic origins.

Reduced uncertainty: Multiple lines of evidence reduced uncertainty in channel identification and characterization.

Improved geological understanding: The integrated approach provided deeper insights into the complex processes that formed and modified the channel systems.

These advantages were particularly valuable for characterizing the complex channel systems in the H Formation, where conventional amplitude-based interpretation often proved inadequate.

Conclusions

This study successfully integrated machine learning techniques with traditional seismic interpretation methods to enhance the identification and characterization of subtle channel features within the H carbonate formations of offshore Abu Dhabi. By combining unsupervised and supervised waveform classification with multi-attribute analysis using Self-Organizing Maps (SOM) and Principal Component Analysis (PCA), we developed a robust workflow for delineating complex channel systems.

The results demonstrate that this integrated approach provides a more comprehensive understanding of channel features compared to conventional amplitude-based interpretation. We were able to not only identify the presence and morphology of these channels but also gain deeper insights into their genetic mechanisms, including the interplay of tectonic influences, sea-level fluctuations, karstification processes, and depositional controls. The visual evidence from waveform classifications, detailed facies maps, and RGB blended sections clearly illustrates the effectiveness of the methodology in revealing intricate subsurface details.

This enhanced understanding of channel systems has significant implications for reservoir characterization and development planning in the H Formation and similar carbonate settings. The ability to accurately map and characterize these features reduces exploration and development risks, leading to optimized drilling strategies and improved hydrocarbon recovery. The methodology presented herein serves as a valuable framework for future studies in challenging geological environments, emphasizing the power

of integrated data analysis and advanced machine learning techniques in unlocking the full potential of seismic data.

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